



Taeniasis — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. Which animal is the intermediate host of *Taenia solium*?

EASY

- A. Cow
- B. Pig
- C. Dog
- D. Sheep
- E. Chicken

2. The infective larval stage acquired by eating undercooked beef is:

EASY

- A. *Cysticercus bovis*
- B. *Cysticercus cellulosae*
- C. Rhabditiform larva
- D. Filariform larva
- E. Hydatid cyst

3. Which tapeworm has a scolex with a rostellum bearing hooks?

EASY

- A. *T. saginata*
- B. *T. solium*
- C. Both species
- D. Neither species
- E. *Enterobius vermicularis*

4. For adult intestinal *Taenia*, which drug is given at 5–10 mg/kg as a single oral dose?

EASY

- A. Pyrantel pamoate
- B. Praziquantel
- C. Levamisole
- D. Thiabendazole
- E. Ivermectin

5. A 38-year-old man notices flat white segments in his underwear, with mild nausea and hunger pains. He often eats undercooked beef. Which stage did he swallow?

MEDIUM

- A. Embryonated egg
- B. Gravid proglottid
- C. Rhabditiform larva
- D. *Cysticercus* in muscle
- E. Filariform larva

6. A woman who has never eaten pork is diagnosed with cysticercosis. Her husband was recently treated for *T. solium* taeniasis. What is the most likely route?

MEDIUM

- A. Eating undercooked beef
- B. Eating undercooked pork
- C. Skin penetration by larvae
- D. Bite of an infected animal
- E. Accidental ingestion of eggs shed by an infected carrier

7. MRI shows non-inflamed cysts in a patient with mild symptoms. Which statement about the viable cysts is correct?

MEDIUM

- A. They trigger intense neutrophil infiltration
- B. They cannot inhibit complement
- C. They evade complement and provoke little inflammation
- D. They stimulate strong cytokine release
- E. They are destroyed by host antibodies

8. Which cyst product is a serine-threonine protease inhibitor that blocks complement activation and cytokine production and interferes with neutrophil function?

MEDIUM

- A. Paramyosin
- B. Sulfated polysaccharide
- C. Glutathione-S-transferase
- D. Fc receptor
- E. Taeniastatin

9. Paramyosin helps the cyst evade complement mainly by inhibiting the activity of:

MEDIUM

- A. C3 convertase
- B. C5a
- C. C1q
- D. Factor B
- E. C9

10. Which investigation is considered the most sensitive for detecting neurocysticercosis?

MEDIUM

- A. Stool microscopy for eggs
- B. Serum eosinophil count
- C. Chest X-ray
- D. Skin-prick testing
- E. MRI or CT scan

11. A meat inspector wants a measure that inactivates cysticerci in pork. Which is supported by the lecture?

MEDIUM

- A. Refrigeration at 4 °C overnight
- B. Brief rinsing under running water
- C. Salting for one hour
- D. Cooking to lukewarm (40 °C)
- E. Cooking above 56 °C

12. A 45-year-old has painless 10-15 mm subcutaneous nodules. Which is most accurate?

MEDIUM

- A. Always cause severe pain
- B. Always progress to neurocysticercosis
- C. Usually asymptomatic; pain and swelling appear when cysts are destroyed
- D. Contain adult worms
- E. Are seen only in the eye

13. In some endemic areas *T. solium* causes about 30% of epilepsy. Which epidemiological factor is emphasised?

MEDIUM

- A. Consumption of raw fish
- B. Contaminated drinking water only
- C. Mosquito exposure
- D. People and roaming pigs living in close proximity
- E. Barefoot walking on soil

14. A patient with cysticercosis complains of eye pain and blurred vision. Which statement is correct?

MEDIUM

- A. Ocular disease is caused by the adult worm
- B. Larval cysts in the eye can cause partial or complete loss of vision
- C. Ocular cysticercosis is always asymptomatic
- D. It is caused by *T. saginata* larvae
- E. It resolves without inflammation

15. A man with an adult *T. solium* in his gut asks whether he is otherwise at risk. Which statement is most accurate?

HARD

- A. He can develop cysticercosis through autoinfection
- B. No risk, since only larvae infect tissues via meat
- C. Risk arises only if he eats more pork
- D. He risks only beef-tapeworm disease
- E. Autoinfection occurs only in pigs

16. A patient passes proglottids and also has calcified nodules in thigh muscle. Which single species best explains both findings?

HARD

- A. *T. solium*
- B. *T. saginata*
- C. *T. saginata* with a second infection by beef eggs
- D. *Enterobius vermicularis*
- E. *Ancylostoma duodenale*

17. The metacestode binds host antibodies through Fc receptors. What is the consequence for the parasite?

HARD

- A. The antibodies neutralise the cyst
- B. Complement is amplified on the cyst
- C. Neutrophils are recruited to the cyst
- D. Antibodies are degraded and used as a source of amino acids
- E. The cyst is opsonised for killing

18. A patient with seizures and active parenchymal cysts is started on an antiparasitic drug. What is the lecture-supported regimen?

HARD

- A. Niclosamide alone
- B. Anticonvulsants alone
- C. Praziquantel or albendazole plus anti-inflammatory and anticonvulsant drugs
- D. Single-dose praziquantel alone
- E. Antibiotics plus steroids only

19. Symptoms often worsen after cysticidal therapy begins. Which lecture statement best explains this?

HARD

- A. Destruction of the parasite induces strong inflammation, granuloma formation and fibrosis
- B. The drug spreads eggs to new sites
- C. Viable cysts are highly immunogenic
- D. The adult worm is released into the brain
- E. Complement is inhibited by the drug

20. A village lacks latrines and pigs roam freely. Which step of the life cycle keeps transmission going?

HARD

- A. Humans eat undercooked beef cysticerci from pigs
- B. Larvae penetrate bare feet
- C. Mosquitoes carry eggs between pigs
- D. Pigs eat eggs voided in human feces and humans later eat undercooked pork
- E. Filariform larvae infect pigs in soil

✓ Answer key

Q	Answer	Difficulty	Why
1	B	EASY	The pig hosts the cysticerci of <i>T. solium</i> (<i>C. cellulosae</i>); the human is the definitive host.
2	A	EASY	Beef carries <i>C. bovis</i> (<i>T. saginata</i>); pork carries <i>C. cellulosae</i> .
3	B	EASY	The lecture figure labels a rostellum with hooks only on <i>T. solium</i> ; <i>T. saginata</i> shows suckers only.
4	B	EASY	Praziquantel is the first drug listed for taeniasis.
5	D	MEDIUM	Taeniasis follows eating meat cysticerci; passed proglottids confirm the adult worm.
6	E	MEDIUM	Cysticercosis is acquired by accidental egg ingestion or autoinfection, not by eating meat.
7	C	MEDIUM	Viable cysts induce little or no inflammation; symptoms mostly follow parasite destruction.
8	E	MEDIUM	Taeniastatin is the protease inhibitor; paramyosin targets C1q.
9	C	MEDIUM	The lecture states paramyosin impairs complement primarily through C1q inhibition.
10	E	MEDIUM	MRI and CT are the most sensitive methods; stool tests find the adult worm, not brain cysts.
11	E	MEDIUM	Cook beef and pork above 56 °C, or freeze at –10 °C for 10 days.
12	C	MEDIUM	Muscle and subcutaneous cysticerci are usually silent; destruction provokes inflammation.
13	D	MEDIUM	Pigs eat eggs from human feces; humans then eat pork or swallow eggs.
14	B	MEDIUM	Ocular cysticercosis causes pain, blurriness and partial or complete visual loss.
15	A	HARD	Cysticercosis is acquired by accidental egg ingestion or autoinfection.
16	A	HARD	Only <i>T. solium</i> has a human tissue larval stage, alongside its adult intestinal form.
17	D	HARD	The parasite subverts antibody binding into nutrition instead of immune killing.
18	C	HARD	Cysticercosis therapy adds anti-inflammatory and anticonvulsant drugs to the antiparasitic.
19	A	HARD	Inflammatory response to dying parasites drives symptoms — hence anti-inflammatory cover. (This link is an inference from the slides.)
20	D	HARD	Eggs voided in feces are ingested by the intermediate host; undercooked meat then completes the cycle.